Artificial Intelligence:

It is branch of computer science that deals with tasks that require human intelligence such as problem solving, planning, decision making, knowledge, reasoning, finding patterns organizing objects, understanding language and many more.

Ex: google maps uses AI to find best route b/w source and destination.

Objectives of AI:

The main objectives are:

(i) Problem solving:

AI should be able to solve complex problems and find best possible solns for it.

(ii) Learning: AI should be able to learn from its previous experience (mistakes), and also learn from data. So it improves performance and accuracy.

(iii) Reasoning and Decision Making:

→ The system should use available info to draw conclusions (Reasoning), and should also be able to make the appropriate desicions (desicion making) from different possib

(iv) Planning: AI should be able to determine a sequence of actions needed to acheive a goal.

(v) Natural Language Understanding: AI should be able to understand and communicate in human language.

Applications:

→ Medical field (Disease diagnosis)
→ Education
→ Banking (fraud detection)
→ Robotics
→ Research
→ Games
→ Languages

AI problems:

o An AI problem is a problem that requires intelligent desicion making, reasoning, searching, planning, learning or understanding to find a solu

Types of problems.

1) Search Problems
2) Planning Problems
3) Classification problems.
4) Prediction problem.

1) Search Problem:

→ searches through all possible solutions.
Ex: DFS, BFS, A^*, AO^*

2) Planning Problem:

→ Planing a sequence of steps to reach goal state.
Ex: developing a project.

3) Classification Problem:

→ The system needs to place something into a category.

4) Prediction: predicts outcome on available info.
Problem → Initial state → Goal state → possible action
→ search solutions → select suitable soln.

AI Characteristics:

- Larage search space: large no of possible solns.
- Knowledge requirement: requires lot of knowledge
Better knowledge better solutions.
- Uncertainity: Sometimes complete info is not available so we might not get proper solution.
- Need for reasoning: should think logically instead of blindly following instructions.

AI techniques:

- Planning
- Reasoning
- Knowledge Representation
- Searching
- Learning.

(2) State Space Search:

It is used to solve problems by exploring all possible states.

→ A state space is the set of all possible states

→ To solve a problem using state space we

(i) Define the problem precisely.

(ii) Analyze the problem

(iii) Identify possible solutions

(iv) Find Best possible solution.

Ex: Tic-Tac Toe, 8 puzzle problem, water jug problem.

4 tuple (S, I, O, G)

S → States O → operators

I → Initial state G → goal state

Ex1:
Tic Tac Toe

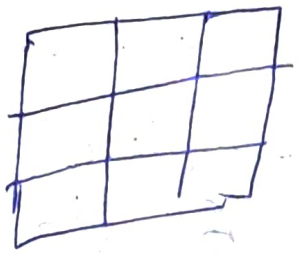
Tic Tac Toe has limited no of possible moves, and has limited blocks.

→ Its moves can be represented.

→ Each move changes the current board.

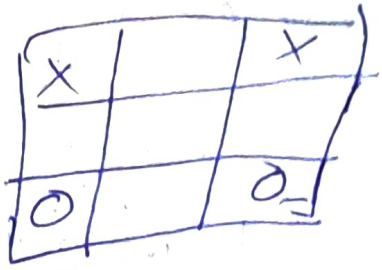
→ Eventually reaches the goal state.

Initial State:

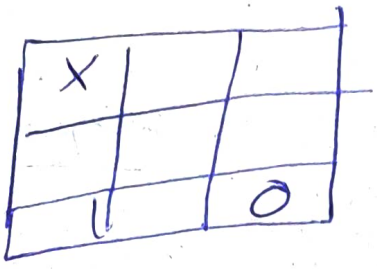


operators X, O, Empty slot.

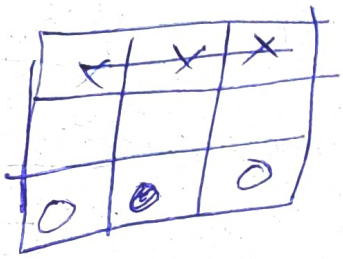
State1:



Int state1:



~~state2~~:
State3:



(Final).
Goal

The game ends when:
- all 3 X's or O's in horizontal or diagonal or vertical direction
- or when there is a tie (all blocks filled but no alignments).

8 puzzle problem:

it is a 3*3 board 1 tile is empty we can move a tile up or down left, right based up available space / moves.
* numbered from 1 to 8

4 tuple (S, I, O, G)
S → set of states
I → Initial state
O → Operators
G → goal state.

Initial state:

1	2	3
7		4
6	8	5

goal state:

1	2	3
8		4
7	6	5

Step 1:

1	2	3
7	8	4
6		5

step 2:

1	2	3
7	8	4

1	2	3
	7	4
6	8	5

step 2:

1	2	3
7		

1	2	3
7	8	4
6		5

1	2	3
7	8	4
	6	5

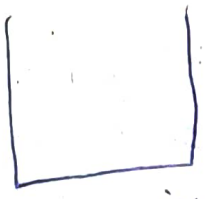
1	2	3
8	8	4
7	6	5

1	2	3
8		4
7	6	5

Water Jug problem:

→ used to explain State space search.

Ex: Two jugs of capacities 4 liters and 3 liters are given. The aim is to measure exactly 2 liters of water using jugs



4L jug



3L jug

(x, y) x = amount of water in 4L jug
y = — — — 3L jug.

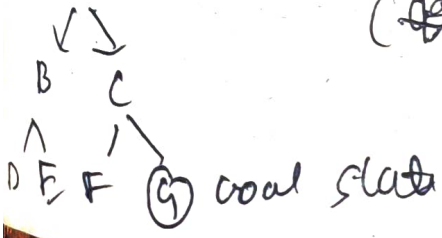
Initial state (0, 0).
Goal state: Any state in which x = 2, or y = 2.

Operations:
→ Fill a jug completely
→ Empty a jug completely
→ Pour water from one jug to other till the first jug is empty or second is full.

(0, 0)
↓
(4, 0)
Fill 4L jug
↓
(1, 3)
pour (nt (3L jug)
↓
(1, 0)
Empty 3L jug
↓
(0, 1)
pour remains wat.

(0, 0)
↓
(0, 3)
↓
(3, 0)
↓
(3, 3)
(4, 2).
(0, 2).
(2, 0) ✓

A Initial



G coal state

(~~Tree Dep~~). graph (or) Tree dep

Production System:

Production System is a method of representing knowledge and solve problems using a set of rules.
These rules are usually written in the form:

IF cond → THEN action.

Ex:
If it is raining → THEN carry umbrella.

Production systems are used in expert systems, problem solving, decision making.

Components of a Production System:

1) Production Rules:

Production Rules are some rules which tells ~~AI~~ what to do based on condition.

IF condition THEN execute statement.

Ex: IF temperature is high → THEN turn on fan.

Rules:
- cond
- Action

2) Working Memory: Working memory stores current information or facts about the problem

Ex: TEMP = 35°C
Fan = ON

The system uses the facts to decide which rule to be applied.

3) Control Strategies: The control strategies are strategies used to select the problem for given problem statement when there are several rules that can be applied for the same problem statement.

a) Rule ordering: Rules are given specific priority order and higher priority rules are selected first.

b) Specificity: choose rules that are more specific conditions. Ex: A rule matching 3 conditions is preferred over matching 1 cond.

c) Recency: prefer the most recently updated or added rule.

d) Conflict Resolution: The multiple rules can be applied to the same problem we can use control strategies such as rule ordering, specificity, recency.

Categories of Production System:

Monotic Production system:

In this system applying a rule doesn't change or remove existing facts. New facts are added but old facts remains same.

Ex: Ram is student
Ram is intelligent.

The first fact is still true.

Non-Monotonic Production System:

→ Applying rule doesn't effect existing facts. ↳ remove or change.

Normally, birds can fly
Penguin is a bird
Penguin cannot fly

→ Normally, birds can fly except penguin.
but penguin cannot fly.

Partially - - - -

→ order of applying rules doesn't matter but all rules to be applied.

Ex: Move A → Move B → Move C }
Move B → Move A → Move C }

Adv:

→ Easy to understand
- - - - Implement
→ Applied in expert sys, decision making, problem
→ Reusable.

Control Strategies:

→ Decides which rule should be selected and updated. When there are several rules that can be applied for the same problem.

main steps:

working memory
↓
Find application Rules
↓
Select a rule
↓
Apply a rule
↓
update working memory
↓
Repeat.

Charactes:

Rule based
→ Seperates knowledge from control ~~strates~~
→ problem solving
→ easy to use and update.

Ex: Sky is clouds
Sky is clouds
↓
It may rain.
↓
Carry umbrella.

It may rain
↓
carry umbrella

Types of control strategies:

1) Forward Reasoning: Start with known facts and applies rules to drive new facts until goal state is reached. (data driven)

Facts → Rules → New facts → Goal

2) Backward Reasoning: Start with goal and works backward to find required facts to prove the goal.

Goal → Rules → required Facts.

Ex: Goal → Carry Umbrella

Carry Umbrella → It may rain

It may rain → (Cloudy weather).

3) Selection of Rules:

Rule selection is based on

→ Priority
→ specificity (- from prev question)
→ Recency
→ Rule ordering.

4) Conflict resolution: When multiple rules can be applied for same problem there are chances of conflicts for resolving conflicts we can.

Ex: Facts: Fever
cough

R1: IF fever THEN take medicine
R2: IF fever and cough THEN consult doctor
Both rules applicable.

Methods of conflict resolution:
- priority
- specificity
- recency.

Uniformed search BFS and DFS:

Uninformed search is an AI search technique that searches for a solution without using heuristic or additional knowledge about the problem. It uses only info given in problem.

Two search methods are:
1) Breadth First search (BFS)
2) Depth First Search (DFS)

1) Breath First Search (BFS):

BFS explores the search tree level by level. It visits all nodes at one level before moving to next level.

working:

→ Start from initial node.
→ Visit all its immediate children.
→ Then visit the children of those nodes.

- Continue till goal state is reached.
- BFS uses a queue (FIFO).

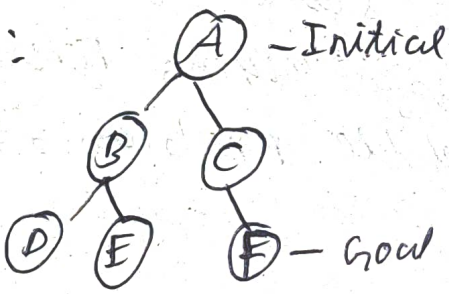
Algorithm:

1) Insert the initial node into the queue.
2) Remove the first node from queue.
3) Check whether it is the goal.
4) If not add its unvisited children to the queue.
5) Repeat till goal node is found or queue is empty.



A → B → C → D → E → F

Ex:



Advantages:

- finds shortest path.
- complete if a soln exists.
- easy to understand and implement

Disadvantages:

- Requires more memory.
- Can be slow for large search spaces.
- stores many node in the queue.

2) Depth First search (DFS):

- Depth first serach explores a path as deep as possible before backtracking and trying another path.

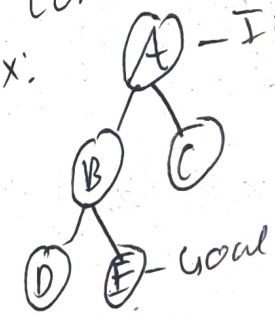
-> ~~Select one child~~
working:

-> Start from the initial node.
-> Select one child and continue deeper.
-> when no further child is available, backtrack.
-> continue till goal is found.
-> DFS uses stack (LIFO).

Algo:

-> Insert initial node into stack
-> Remove top node.
-> check whether it is the goal.
-> If not add its unvisited children to the stack.
-> continue till goal is found or stack is empty.

Ex:



Initial

goal = E

A → B → D → E.



Advantages:

-> less memory.
-> simple to implement.
-> useful when soln is deep in search tree.

Disadvantages:

- Doesn't always guarantee shortest path.
- can go very deep in unproductive path.
- May not find a soln in infinite search space.

BFS	DFS
→ level by level search	→ dept wise search
→ uses queue	→ uses stack.
→ FIFO	→ LIFO
→ shortest path for equal costs	→ shortest path not guar
→ memory	less memory

Heuristic Search :-

problem characteristics,

remaing refer from pdf.